Def.
A group $D_{2n} \overset{\text{def}}{=} \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$ is the dihedral group of order $2n$.

Prop. Let D_{2n} be the dihedral of order $2n$.
1). For any $x \in D_{2n} \setminus \langle r \rangle$, $rx = xr^{-1}$
2). For any $x \in D_{2n} \setminus \langle r \rangle$, $|x| = 2$.
3). $D_{2n} = \langle s, sr \rangle$.
4). If $n = 2k$ is even with $n \geq 4$, then r^k has order 2 and $\langle r^k \rangle = \{1, r^k\} = Z(D_{2n})$.
5). If n is odd with $n \geq 3$, then $Z(G) = \{1\}$.

Proof. Let $x \in D_{2n}$ be given. Then $x = r^a s$ for some $a = 0, 1, \cdots, n-1$, by $x \notin \langle r \rangle$,
So, $rx = r^{a+1} s = r^a \cdot r \cdot s = r^a s r^{-1} = x \cdot r^{-1}$.
And, $x^2 = r^a s r^a s = r^{a-1} s r^{-1} r^a s = \cdots = s r^{-a} \cdot r^a \cdot s = s^2 = 1$, and $x \neq 1$,
so $|x| = 2$. ∎ The 3) is given by $\langle s, sr \rangle \subseteq D_{2n}$ and $ssr = r \in \langle s, sr \rangle$, So
$\langle s, sr \rangle \geq \langle s, r \rangle = D_{2n}$. ∎
To determine the Center of D_{2n}, observe that: given $a \in \{0, 1, \cdots, n-1\}$,

$r^a \in Z(D_{2n}) \Leftrightarrow r^a s = s r^a = s r^{-a} \Leftrightarrow r^{2a} = 1$, by the cancelation law.

Since $r^{2a} = 1$ if and only if $n \mid 2a$, so $r^a \in Z(D_{2n})$ if and only if

$$a = \begin{cases} 0 & \text{if } n \text{ is odd} \\ 0, \frac{n}{2} & \text{if } n \text{ is even.} \end{cases}$$

And, given $x \in D_{2n} \setminus \langle r \rangle$, $x \in Z(D_{2n})$ implies that $xr = rx = xr^{-1} \Rightarrow r^2 = 1$.
But, it contradicts to $|r| = n \geq 3$. Consequently, for $n \geq 3$,

$$Z(G) = \begin{cases} \{1\} & \text{if } n \text{ is odd} \\ \{1, r^k\} & \text{if } n = 2k \text{ is even} \end{cases}$$

∎

Prop. If n is even, then $D_{2n} \cong \langle (1\ 2\ \cdots\ n), (2\ n)(3\ n-1)\cdots(\frac{n}{2}\ \frac{n}{2}+2) \rangle \leq S_n$
If n is odd, then $D_{2n} \cong \langle (1\ 2\ \cdots\ n), (2\ n)(3\ n-1)\cdots(\frac{n+1}{2}\ \frac{n+1}{2}+1) \rangle \leq S_n$.

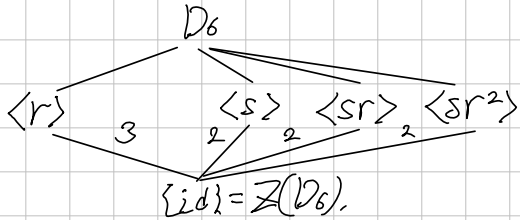
Proof In case of n is even number, observe that

$(1\ 2\ \cdots\ n)(2\ n)(3\ n-1)\cdots(\frac{n}{2}\ \frac{n}{2}+2)(1\ 2\ \cdots\ n)$
$= (1)(2\ n)(3\ n-1)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$

Moreover, $(1\ 2\ \cdots\ n)^n = id$ is clear, and since $(2\ n)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$ is product of disjoints two-cycles, so it has order 2.
So, corresponding to $r = (1\ 2\ \cdots\ n)$ and $s = (2\ n)\cdots(\frac{n}{2}\ \frac{n}{2}+2)$,
it satisfies the representation of the definition. Similarly, n is odd, can shown. ☐

D_6.

By the theorem above, $D_6 \cong S_3$. Now, explicitly,

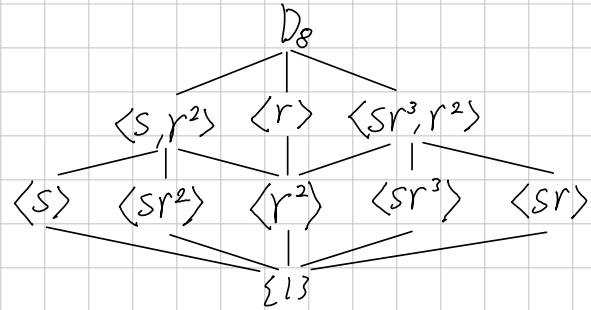


Normalizer:

$N_{D_6}(\langle r \rangle) = D_6$

D_8

$D_8 = \langle r, s \mid r^4 = s^2 = 1,\ rs = sr^{-1} \rangle$



Normalizer: Since index of 2 subgroups must be normal,
the normalizer of $\langle s, r^2 \rangle$, $\langle r \rangle$, $\langle sr^3, r^2 \rangle$ is just D_8.
$N_{D_8}(\langle s \rangle) = \langle s, r^2 \rangle$, because: since $|\langle s, r^2 \rangle : \langle s \rangle| = 2$, $\langle s, r^2 \rangle \leq N_{D_8}(\langle s \rangle)$, but

$$r\langle s \rangle r^{-1} = \{1, sr^2\} \neq \langle s \rangle, \text{ so } N_{D_8}(\langle s \rangle) \neq D_{2n}.$$

Similarly $N_{D_8}(\langle sr^2 \rangle) = \langle s, r^2 \rangle$. But, $N_{D_8}(\langle r^2 \rangle) = D_8$, because:
First way: since $|\langle s, r^2 \rangle : \langle r^2 \rangle| = |\langle r \rangle : \langle r^2 \rangle| = 2$, therefore

$$\langle s, r^2 \rangle, \langle r \rangle \subseteq N_{D_8}(\langle r \rangle), \text{ so } N_{D_8}(\langle r \rangle) = D_8.$$

Alternatively, since $sr^2s^{-1} = sr^2s = r^{-2}s \cdot s = r^2$, so $s, r \in N_{D_8}(\langle r \rangle)$,
so $\langle s, r \rangle \subseteq N_{D_8}(\langle r \rangle)$. ($r \in D_8$ clearly normalizes $\langle r^2 \rangle$).

Centralizer: First, $C_{D_8}(D_8) = \{1, r^2\} = \langle r^2 \rangle$ because: $sr^2 = r^{-2}s$ and
clearly r^2 commutes all elements in $\langle r \rangle$, so r^2 commutes all elements of D_8.
but, $sr^3 = rs \neq r^3s$, so $r, s \notin Z(G)$ and $sr^2 \notin Z(G)$ because
$sr^2 \in Z(G) \Rightarrow sr^2r^{-2} = s \in Z(G)$ since the center is a subgroup.
Moreover, $sr \cdot s = r^3ss = r^3 \neq ssr = r$, so $sr \notin Z(G)$. Thus $Z(G) = \langle r \rangle$
Next, $C_{D_8}(\langle r \rangle) = \langle r \rangle$, because: since $rs = sr^3 \neq sr$, so $s \notin C_{D_8}(\langle r \rangle)$ and
clearly all elements in $\langle r \rangle$ commute with the all elements in $\langle r \rangle$, so
$\langle r \rangle \subseteq C_{D_8}(\langle r \rangle) \subsetneq D_8$, so the Lagrange's theorem gives the result.

Commutator: Since the derived set is actually normal subgroup and
it induces the largest abelian quotient, so $D_8' = \langle r^2 \rangle$, because $D_8/\langle r^2 \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2$,
because it contains two elements order 2, but it is not cyclic.

ΓD10